PRACTICAL NO : 5

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ROLL NO : TCC71

Input File : TCC71.txt

Hey, I just wanted to check if you’re available to meet tomorrow afternoon for a quick discussion about the project.

WC_Mapper.java

```
  package com.javatpoint;
   import java.io.IOException;
   import java.util.StringTokenizer;
   import org.apache.hadoop.io.IntWritable;
   import org.apache.hadoop.io.LongWritable;
   import org.apache.hadoop.io.Text;
   import org.apache.hadoop.mapred.MapReduceBase;
   import org.apache.hadoop.mapred.Mapper;
   import org.apache.hadoop.mapred.OutputCollector;
   import org.apache.hadoop.mapred.Reporter;
   public class WC_Mapper extends MapReduceBase implements
Mapper<LongWritable,Text,Text,IntWritable>{
       private final static IntWritable one = new IntWritable(1);
       private Text word = new Text();
       public void map(LongWritable key, Text value,OutputCollector<Text,IntWritable>
output,
             Reporter reporter) throws IOException{
           String line = value.toString();
           StringTokenizer  tokenizer = new StringTokenizer(line);
           while (tokenizer.hasMoreTokens()){
               word.set(tokenizer.nextToken());
               output.collect(word, one);
           }
       } }
```

WC_Reducer.java

```
package com.javatpoint;
import java.io.IOException;
import java.util.Iterator;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WC_Reducer  extends MapReduceBase implements
Reducer<Text,IntWritable,Text,IntWritable> {
public void reduce(Text key, Iterator<IntWritable>
values,OutputCollector<Text,IntWritable> output,
 Reporter reporter) throws IOException {
int sum=0;
while (values.hasNext()) {
sum+=values.next().get();
}
output.collect(key,new IntWritable(sum));
} }
```

WC_Runner.java

```
package com.javatpoint;
import java.io.IOException;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.mapred.TextInputFormat;
import org.apache.hadoop.mapred.TextOutputFormat;
public class WC_Runner {
```

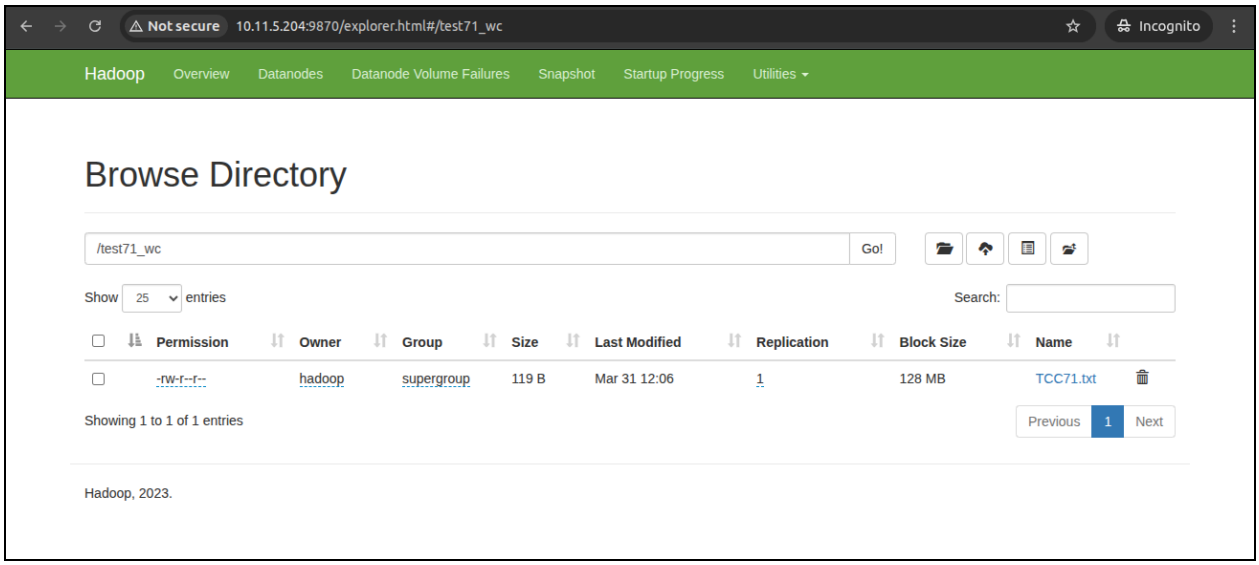
```
    public static void main(String[] args) throws IOException{
        JobConf conf = new JobConf(WC_Runner.class);
        conf.setJobName("WordCount");
        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);
        conf.setMapperClass(WC_Mapper.class);
        conf.setCombinerClass(WC_Reducer.class);
        conf.setReducerClass(WC_Reducer.class);
        conf.setInputFormat(TextInputFormat.class);
        conf.setOutputFormat(TextOutputFormat.class);
        FileInputFormat.setInputPaths(conf,new Path(args[0]));
        FileOutputFormat.setOutputPath(conf,new Path(args[1]));
        JobClient.runJob(conf);
    }
}
```

TERMINAL COMMANDS :

```
hadoop@student:~/hadoop$ java --version
openjdk 11.0.30 2026-01-20
OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu122.04, mixed mode, sharing)
hadoop@student:~/hadoop$ su - hadoop
Password:
hadoop@student:~$ cd hadoop
hadoop@student:~/hadoop$ hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /home/hadoop/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
hadoop@student:~/hadoop$ nano TCC71.txt
hadoop@student:~/hadoop$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [student]
Starting resourcemanager
Starting nodemanagers
hadoop@student:~/hadoop$ hdfs dfs -mkdir /test71_wc
hadoop@student:~/hadoop$ hdfs dfs -put admin:///home/hadoop/hadoop/TCC71.txt /test71_wc
```

```
hadoop@student:~/hadoop$ ifconfig
eno1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.11.5.204  netmask 255.255.0.0  broadcast 10.11.255.255
        inet6 fe80::a709:7771:4c7:e8e1  prefixlen 64  scopeid 0x20<link>
        ether 50:65:f3:2e:5a:fd  txqueuelen 1000  (Ethernet)
        RX packets 79872  bytes 64801108 (64.8 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 39701  bytes 8097189 (8.0 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 20  memory 0xf7c00000-f7c20000

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 2117  bytes 259929 (259.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2117  bytes 259929 (259.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```




```
hadoop@student:~/hadoop$ nano WC_Mapper.java
hadoop@student:~/hadoop$ nano WC_Reducer.java
hadoop@student:~/hadoop$ nano WC_Runner.java
hadoop@student:~/hadoop$ javac -classpath "$(hadoop classpath)" -d . WC_Mapper.java WC_Reducer.java WC_Runner.java
hadoop@student:~/hadoop$ jar -cvf wordcount71.jar com
added manifest
adding: com/(in = 0) (out= 0)(stored 0%)
adding: com/javatpoint/(in = 0) (out= 0)(stored 0%)
adding: com/javatpoint/WC_Mapper.class(in = 1874) (out= 768)(deflated 59%)
adding: com/javatpoint/WC_Reducer.class(in = 1546) (out= 613)(deflated 60%)
adding: com/javatpoint/WC_Runner.class(in = 1485) (out= 728)(deflated 50%)
```

```
hadoop@student:~/hadoop$ hadoop jar /home/hadoop/hadoop/wordcount71.jar com.javatpoint.WC_Runner /test71_wc/TCC71.txt /r_output71
2026-03-31 12:31:15,497 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-31 12:31:15,569 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-31 12:31:15,569 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-31 12:31:15,580 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-03-31 12:31:15,692 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and
execute your application with ToolRunner to remedy this.
2026-03-31 12:31:15,793 INFO mapred.FileInputFormat: Total input files to process : 1
2026-03-31 12:31:15,929 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-31 12:31:16,472 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1173693674_0001
2026-03-31 12:31:16,472 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-31 12:31:16,656 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-03-31 12:31:16,658 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-31 12:31:16,658 INFO mapreduce.Job: Running job: job_local1173693674_0001
2026-03-31 12:31:16,660 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-03-31 12:31:16,678 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-31 12:31:16,678 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, igno
re cleanup failures: false
2026-03-31 12:31:16,783 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-03-31 12:31:16,785 INFO mapred.LocalJobRunner: Starting task: attempt_local1173693674_0001_m_000000_0
2026-03-31 12:31:16,847 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-31 12:31:16,847 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, igno
re cleanup failures: false
2026-03-31 12:31:17,043 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-03-31 12:31:17,074 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/test71_wc/TCC71.txt:0+119
2026-03-31 12:31:17,130 INFO mapred.MapTask: numReduceTasks: 1
2026-03-31 12:31:17,283 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-03-31 12:31:17,284 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-03-31 12:31:17,284 INFO mapred.MapTask: soft limit at 83886080
2026-03-31 12:31:17,284 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-03-31 12:31:17,284 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-03-31 12:31:17,289 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-03-31 12:31:17,688 INFO mapreduce.Job: Job job_local1173693674_0001 running in uber mode : false
2026-03-31 12:31:17,689 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-31 12:31:18,575 INFO mapred.LocalJobRunner:
```

OUTPUT

```
hadoop@student:~/hadoop$ hdfs dfs -cat /r_output71/part-00000
Hey,    1
I       1
a       1
about   1
afternoon       1
available       1
check   1
discussion      1
for     1
if      1
just    1
meet    1
project.        1
quick   1
the     1
to      2
tomorrow        1
wanted  1
you’re  1
hadoop@student:~/hadoop$
```